

Exit Report of Project FinovistaAI for Customer Israel Discount Bank, Ltd.

Customer: Israel Discount Bank, Ltd.

Team Members: Lior Abuhav, Niv Vaknin, Limor Hodory, Liav Katry

1. Overview

FinovistaAI set out to address key challenges in wealth management and investment advisory by leveraging advanced language models. Specifically, this project combined **Prompt Optimization** (via MIPROv2), **Chain-of-Thought (CoT)** reasoning extracted from a larger model (GPT), and a **smaller, cost-effective LLM** (Falcon 7B). The end goal was to deliver **faithful, context-rich** financial insights at scale without incurring the prohibitive costs and complexities of large-scale model fine-tuning.

Our solution focuses on bridging two gaps:

1. **Vague User Queries** in finance, where customers may lack the precise financial terminology or context to formulate well-defined questions.
2. **Cost and Performance Trade-offs** of using large-scale language models in enterprise settings.

By integrating a Retrieval-Augmented Generation (RAG) approach with an optimized Chain-of-Thought program, FinovistaAI enables **Israel Discount Bank** to provide high-quality, trustworthy financial guidance—at scale and with reduced operational costs.

2. Business Domain

Industry: Wealth Management & Investment Advisory

Context:

- Traditional banks and financial institutions like Israel Discount Bank are increasingly incorporating AI-driven advisory tools to better serve retail clients, private banking customers, and institutional investors.
- The demand for **scalable, data-driven** insights is growing, especially in areas like portfolio analysis, risk assessment, regulatory compliance, and personalized investment strategies.

Key Domain Characteristics:

- **Financial Complexity:** Advisors deal with large volumes of data (e.g., market data, regulatory guidelines, client information).
- **Trust & Compliance:** Accurate and transparent advice is crucial due to regulatory and reputational risks.
- **Cost Constraints:** Finetuning or training large AI models (like GPT-scale) for specialized tasks can be prohibitively expensive.

3. Business Problem

3.1 Ambiguous User Queries

- **Issue:** Many users do not have specialized financial knowledge, leading to incomplete or vague investment-related questions.
- **Impact:** AI-driven advisors can deliver suboptimal or imprecise recommendations, undermining trust and user satisfaction.
- **Relevant KPI:**
 - *Query Clarity Index:* Frequency of user queries that require additional clarification.

3.2 High Cost of Model Fine-Tuning

- **Issue:** Large-scale model fine-tuning (e.g., GPT-level) can be cost-prohibitive, both in compute resources and compliance overhead.
- **Impact:** Firms either avoid advanced personalization or rely on smaller, less capable models.
- **Relevant KPI:**
 - *Model Training Cost:* Overall cost of hardware/software for model training.

3.3 Performance Gaps in Smaller Models

- **Issue:** While smaller models are cheaper and easier to deploy, they typically fall short in providing context-rich, high-fidelity insights.
- **Impact:** Users receive less accurate or less relevant recommendations, leading to potential financial losses or lower customer satisfaction.
- **Relevant KPI:**
 - *Accuracy Score:* Quality and correctness of financial insights as measured against expert or market benchmarks.

4. Data Processing

4.1 Original Dataset Schema

1. Internal Dataset

- A set of domain-specific Q&A examples from Israel Discount Bank, focusing on typical retail investment questions, compliance guidelines, and market analysis.

2. Synthetic Dataset

- To supplement limited internal data, we generate additional Q&A pairs using a larger LLM (GPT) in a **few-shot** manner.
- These samples simulated varied financial scenarios (portfolio diversification, risk management, macroeconomic insights) based on given scenarios from Israel Discount Bank internal dataset.

Core Fields in each Q&A record included:

- *About_me* - User profile (e.g., age, occupation, investment preferences).
- *Context* - Background information or market trends relevant to the query.
- *Question* - The user's investment query or financial concern.
- *Answer* - The system's response or recommendation.

4.2 Data Processing Workflow

4.2.1 MIPROv2 Optimization

1. **Data Collection:** Merged internal Q&A with synthetic data.
2. **Model fine-tuning:** Use internal Q&A to fine-tune internal Falcon model.
3. **Integration into MIPROv2 Pipeline:** Internal Q&A and synthetic generated Q&A used to **train** and **validate** the optimized Chain-of-Thought prompts, generating an optimized CoT program ensuring faithfulness and context alignment.

4.2.2 RAG Integration

1. **Data Collection:** Gathered internal financial documents, market analyses, and user guides (e.g from Alpaca).
2. **Vector Indexing (for RAG):** Indexed all documents and text passages using Qdrant Vector DB to enable quick, similarity-based retrieval at runtime.

4.3 Final Input Schema for Model

- *User Query*
- *Retrieved Context* (via RAG)
- *Optimized CoT Prompt* (produced by MIPROv2)

5. Validation

5.1 Approach

We evaluated the solution on a test set covering diverse financial scenarios, measuring:

- **Context Relevancy:** How closely answers aligned with retrieved documents and user intent.
- **Context Recall:** How effectively the system retrieved all pertinent information from the knowledge base.
- **Faithfulness:** Ensuring the output was directly supported by the input prompt and context (minimizing hallucinations).

5.2 Baseline vs. Optimized Results

Baseline (Non-optimized Falcon 7B + Basic RAG)

- Context Relevancy: **0.1203**
- Context Recall: **0.1343**
- Faithfulness: **0.3365**

Final System (MIPROv2 CoT + RAG + Falcon 7B)

- Context Relevancy: **0.1239** (+3.0% improvement)
- Context Recall: **0.1543** (+14.9% improvement)
- Faithfulness: **0.484** (+43.8% improvement)

Interpretation:

- The largest gain was in **Faithfulness**, indicating the final system was significantly less likely to invent unsupported facts or deviate from the user’s original prompt.
- Improvements in Relevancy and Recall show the solution’s enhanced ability to incorporate the right external context.

6. Solution Architecture

6.1 High-Level Description

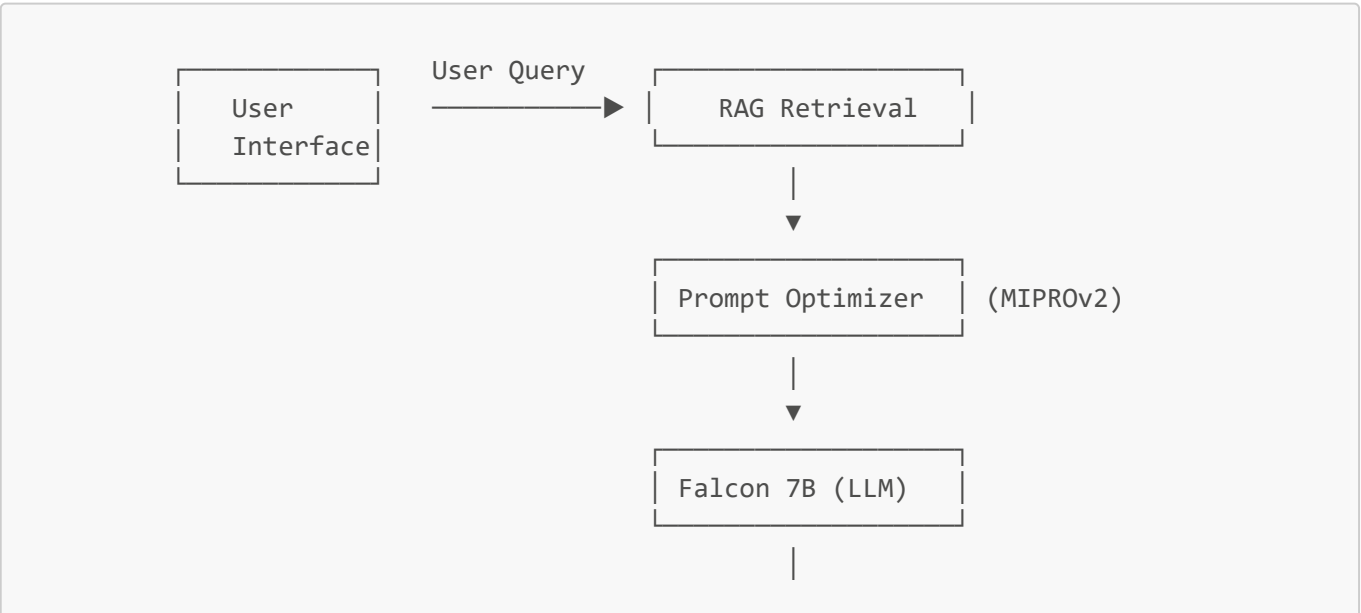
The solution architecture consists of two main components:

- 1. **Prompt Optimizer (MIPROv2)**
 - **Role:** Generates and refines a Chain-of-Thought (CoT) by leveraging a stronger LLM (GPT or similar). Focuses on ensuring that final prompts are “faithful” to the user’s query.
 - **Output:** A structured reasoning program (CoT) that smaller models (e.g., Falcon 7B) can follow to produce more accurate responses.
- 2. **Inference Pipeline**
 - **Steps:**
 - 1. **User Query** → 2. **Document Retrieval via RAG (Qdrant)** → 3. **Prompt Optimization** → 4. **Falcon 7B Inference** → 5. **Final Answer**
 - **Purpose:** Provide relevant context through RAG, apply the optimized Chain-of-Thought, and generate faithful answers at scale.

6.2 Implementation Details

- **RAG Layer (Qdrant):** A vector database storing internal financial documents, market analyses, and user guides.
- **Falcon 7B Model:** A smaller, domain-fine-tuned LLM chosen for its cost-effectiveness.
- **MIPROv2:** Applied prior to runtime to produce a generalizable CoT that each user query can leverage, avoiding expensive large-model fine-tuning.

6.3 Architecture Diagram



▼

Response

7. Benefits

7.1 Company Benefit (Internal Only)

- **Revenue Potential:** Ability to package and sell a cost-effective AI advisory solution to additional financial institutions.
- **Reduced Development Time:** Prompt engineering approach (vs. large-scale model training) significantly shortened solution cycles.
- **Scalable IP:** The MIPROv2-based methodology can be repurposed for future domain expansions (e.g., insurance, retail banking).

7.2 Customer Benefit (Israel Discount Bank)

1. **Cost Efficiency:** By avoiding large-scale model fine-tuning, the bank can deploy advanced AI capabilities with a smaller hardware footprint and lower overhead.
2. **Improved Customer Trust:** Higher faithfulness reduces the likelihood of giving irrelevant or misleading guidance, strengthening user satisfaction and brand reputation.
3. **Faster Time-to-Market:** The solution can be quickly adapted to new products or market conditions without retraining an entire large model.

8. Learnings

8.1 Project Execution

- **Stakeholder Alignment:** Early engagement with Israel Discount Bank's compliance and data teams was essential for ensuring domain relevance and regulatory alignment.
- **Iterative Feedback:** Weekly reviews and agile sprints allowed for rapid iteration on both data processing and prompt optimization.

8.2 Data Science / Engineering

- **Few-Shot Synthetic Data Generation:** Combining real data with GPT-generated synthetic samples effectively expanded the training corpus, especially in underrepresented scenarios.
- **Chain-of-Thought Extraction:** Using a stronger LLM (GPT) to "teach" a smaller model via a structured CoT overcame many constraints of small-model capacity.

8.3 Domain

- **Financial Advisory Nuances:** Small details—like regulatory disclaimers, local market conditions, and product-specific constraints—must be integrated into the AI workflow to maintain credibility and compliance.
- **Context Sensitivity:** Investment recommendations can be risk-adjusted or time-sensitive; the system's ability to retrieve up-to-date or domain-specific context is critical.

8.4 Product

- **MIPROv2 Adaptability:** The tool is model-agnostic and can scale to other business lines within the bank (e.g., mortgage advice, consumer lending) by adjusting the prompt templates accordingly.
- **Retrieval Augmented Generation (RAG) + Vector DB:** Proved essential for indexing large volumes of internal documents, enabling on-demand recall of relevant data.

8.5 What's Unique / Specific Challenges

- **High-Faithfulness Requirement:** Due to regulatory and reputational stakes, the system needed robust controls against hallucinations or irrelevant outputs.
- **Privacy & Compliance:** Ensuring the solution adhered to data protection rules meant that large-scale model fine-tuning offsite was not an option, reinforcing the need for in-house prompt optimization.

9. Next Steps

1. Integration Phase (Q2 2025)

- **Owner:** FinovistaAI & Israel Discount Bank Tech Team
- **Goal:** Migrate the MIPROv2-optimized system into the bank's production environment, ensuring compliance with internal security protocols.

2. Performance Monitoring (Monthly Check-Ins)

- **Owner:** Data Science Team at Israel Discount Bank
- **Goal:** Track improvements in user query resolution rates, faithfulness, and ROI metrics.

3. Expansion to Other Business Units (Q3–Q4 2025)

- **Owner:** Israel Discount Bank Product Management
- **Goal:** Extend the approach to new domains (e.g., mortgage advisory, credit lines, and corporate lending).

4. Online Optimizer Updates (Ongoing)

- **Owner:** FinovistaAI
- **Goal:** Roll out continuous updates for MIPROv2 so it can adapt based on emerging use cases and user feedback, maintaining high fidelity over time.

10. Appendix

A. Additional Materials

1. Baseline System Report (Key Metrics Recap)

- *Baseline Context Relevancy:* 0.1203
- *Baseline Context Recall:* 0.1343
- *Baseline Faithfulness:* 0.3365

2. Final System Report (Key Metrics Recap)

- *Final Context Relevancy*: 0.1239
- *Final Context Recall*: 0.1543
- *Final Faithfulness*: 0.484

3. Sample Chain-of-Thought Program (Excerpt)

```
"instructions": "Given a question, provide a step-by-step reasoning process that leads to the solution, followed by the final answer. Ensure that the reasoning is structured and deliberate, starting with \"Reasoning: Let's think step by step in order to,\" and the final answer is prefixed with \"Answer:\". Both fields should be clearly formatted and directly address the query.",
```

4. Sample Chain-of-Thought Program Output

Original Input Example:

```
{
  "about_me": "I'm a 40-year-old entrepreneur interested in renewable energy.",
  "context": "Solar and wind energy projects are growing globally.",
  "question": "Should I invest in renewable energy companies?"
}
```

Enhanced Input Example:

```
{
  "about_me": "I'm a 40-year-old entrepreneur interested in renewable energy.",
  "context": "Solar and wind energy projects are growing globally. Recent trends show increasing government incentives and rising market demand for renewable solutions. Reasoning: Investments in renewable energy companies are supported by global sustainability goals and advancements in solar and wind technologies. These factors make renewable energy an attractive long-term sector for growth.",
  "question": "Should I invest in renewable energy companies?\nYou can use the following expert reasoning as a reference: 'Renewable energy companies are experiencing significant growth due to increased government support and global sustainability goals. Long-term investments in this sector are considered promising.'"
}
```

5. Support Channels

- **Email:** Support@FinovistaAI.com
- **Support Platform:** FinovistaAI.com (Ticket-based system)

End of Report

We appreciate the collaboration with Israel Discount Bank, Ltd. and look forward to further expanding this solution to meet evolving industry and regulatory demands.